Packet coefficient DFS: active-prefix and restoration audit

6 November 2026. PCC-20261106-d6132b — unverified statement. This packet gives a written source-refinement argument and explicit proof obligations for PacketTensor::entry. No kernel-checked theorem, new algebraic identity, execution result or complexity result is asserted. Every source observation and argument below has unverified producer standing. [pcc:unverified]

The advance is a state correspondence with a remaining-row capacity invariant. It classifies degree rejection as a constant-choice case under that invariant, while retaining the source's defensive mode-undo branch. It separates that same-row site from two committed-parent pop sites, excludes the stale suffix, and relates accepted leaves to ordered row choices. Signs, multiplicities, zero-divisor behavior and creation precision stay explicit. [pcc:unverified]

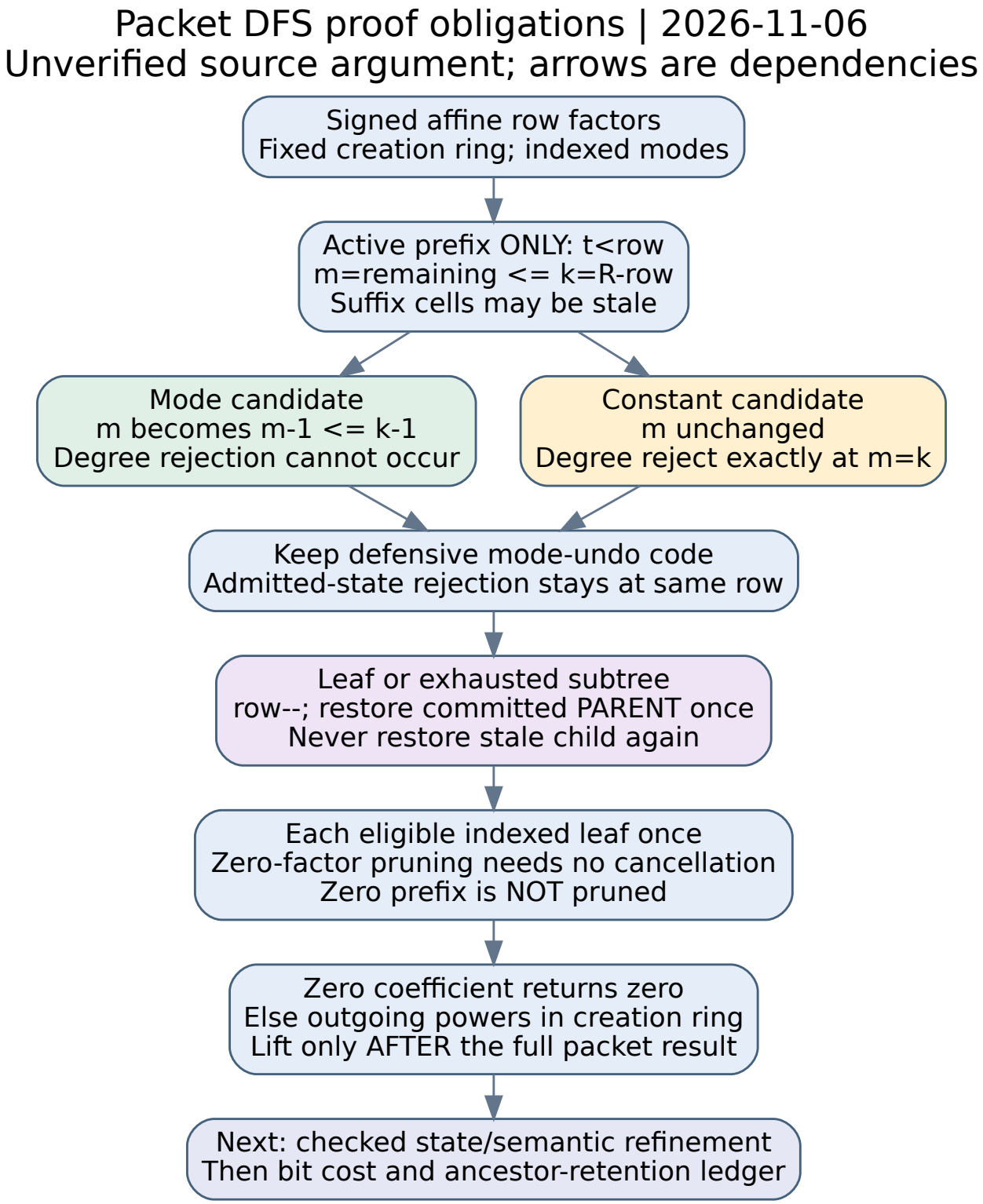


Figure 1: Evidence dependencies; no timing data.

Source and coordination scope

Native source is pinned to crypto 75309f57ba69d7683006108e151a21233a04bb48. The audited entry body matches the Nov5 source pin, while the file now also contains upstream mutation tests; contraction.rs additionally registers the upstream guards module. Whole-file or whole-crate equality is not asserted. sources.json (sources.json) records full-file and extracted-body identities. refinement-map.json (refinement-map.json) links the obligations to exact source lines. [pcc:unverified]

The Nov5 tuple contract (https://github.com/isogeny-labs/crypto/blob/8ada1df4e4f4fb433a37438ef776f29ab0912aef/research/hashed_ecdl_tuple_invariant_20261105/README.md) is an unmerged, unverified candidate from PR3478. Its conflicted branch, ledger and journal remain untouched. This additive continuation uses current main in an isolated worktree, coordinated in PR3478 comment6689804776 before publication. No earlier claim or event is rewritten. [pcc:unverified]

Read OpenAI Appendix D.1/D.4.5-D.4.7, especially D.4.6's row-product coefficient and outgoing-power definition, at PDF path last-change 42907c9e999cc00d6abe7a50bfc1023250edd1b9, blob 0503566e95bc587a4076573a45132d97531b8ba6. Its general entry-work and retention proof remains a separate obligation. Atlas matching at crypto-autoresearcher a97d7c1304033a7863609096d425487cd2e2b31b selects the existing hashed-factor-base tensor audit lead. No new mechanism, novelty clearance or full atlas-delta review. [pcc:unverified]

Sources checked (KB): MCP tools unavailable; repository wrapper scripts/research-kb status exits127, no executable. No KB source IDs, revision/refresh or successful searches are invented. The fallback is the pinned atlas, prior packets and primary source; exact query scope and the gap are recorded in VALIDATION.md (VALIDATION.md).

Audit target and assumptions

Work in the packet's fixed creation ring A, with R rows, C columns and q incoming records. Assume correct commutative-ring operations and faithful zero testing; no integral-domain or cancellation assumption is needed in the argument. The implementation of those operations is not independently proved here. Let beta_j be the stored incoming exponent and E their sum. [pcc:unverified]

For selected column bits x, let l_i be the local row's column sum, and a_ij its stored incoming coefficient. For row bit y_i the prepared coefficients are as follows:

Row bit	Constant c_i	Mode coefficient d_ij
true	l_i	a_ij
false	1-l_i	-a_ij

The existing D.4.6 interpretation supplies the audit target: extract the beta coefficient of the ordered product of affine factors F_i(Z)=c_i+sum_j d_ij*Z_j, then multiply the outgoing linear forms raised to their stored exponents. Incoming records are separate formal variables even if their coefficient vectors agree. This restates the source's semantic target; it is not a new certified polynomial identity. [pcc:unverified]

Shape preconditions: each local row has C coefficients; each incoming list has R; outgoing indices are below C; bit arity is R+C. Require representable R+C, R+1, q+1 and E, and successful allocations. A claimed Ok-return refinement must establish these caller conditions rather than infer them from a slice type or an operation-meter count. [pcc:unverified]

Active-prefix and scan-frontier invariant

For an active depth r, define used_j(r) to count labels j+1 in chosen[k] only for k<r. Label0 chooses the constant. The proposed invariant is:

1. 0 <= r <= R; only chosen[0..r] is committed. Entries at and beyond r may be stale and are excluded from used counts.

2. left_j = beta_j-used_j(r) >= 0, and remaining equals the sum of left, with remaining <= R-r after the initial degree guard.

3. prefix[r] is the ordered product of factors selected by that active prefix; prefix[0] is one. Inactive prefix cells are not authoritative.

4. At a non-leaf, choices[r] is the next untried label under the current committed ancestors. It advances before any rejection, and is reset only when entering a new row under a new ancestor prefix.

5. sum accumulates eligible leaves already closed by this frontier, with each ordered choice vector contributing once. Pruned subtrees contribute zero to the target coefficient for the reasons below.

These are candidate refinement predicates, with the following source-coupled review argument; no checked Rust/usize or Lean model is supplied. [pcc:unverified]

Transition and restoration argument

Source transition	Counter/product change	Frontier argument
Initialize	left=beta, remaining=E, prefix[0]=1, row=0	Empty committed prefix, no closed leaf
Read candidate	choice=choices[row]; increment choices[row]	That label is not retried for this ancestor prefix

Source transition	Counter/product change	Frontier argument
Exhausted exponent	Skip mode label with left_j=0	Additional use would exceed beta_j; no later row can undo that use
Zero factor	Skip without consuming an exponent	Every descendant product has this zero factor, in any ring
Tentative mode	Decrement its left_j and remaining	Candidate contribution is temporarily included
Degree rejection	If remaining>R-row-1, source has a guarded in-place mode undo	Under the capacity invariant, rejection can only be a constant choice; no depth was pushed
Commit candidate	Write prefix[row+1], chosen[row], then increment row; reset next-row choices	Extends the active prefix by one factor
Eligible leaf	If remaining=0 add prefix[R]	Adds exactly this ordered row-choice product
Pop from leaf	Decrement row, restore chosen[row] if nonzero	Removes the now-excluded parent's committed contribution
Pop exhausted row	Same parent restoration after all row labels are exhausted	Does not restore the stale choice at the exhausted depth
Root completion	Break at row0 after empty leaf or exhaustion	No parent exists to restore; closed frontier is final

The degree-rejection site is not a stack pop: row stays fixed and the trial never enters chosen[0..r]. The guarded mode undo there is retained even though the capacity argument below makes it unreachable on admitted states. Conversely, after a leaf or exhausted subtree the source decrements row first and restores that parent's choice. The old chosen entry at the current frontier is harmless because the invariant excludes it. Restoring a speculative trial again as a committed child would double-count the undo; the source's control paths keep those cases separate. [pcc:unverified]

For a committed mode choice, used_j increases by one and left_j decreases by one; after the matching pop both changes reverse. Constant choices change neither exponent counter. Prefix multiplication records the same selected factor as chosen, and every later sibling overwrites its own prefix cell. Together with the persistent next-label cursor, this is the proposed local simulation relation between the imperative arrays and a row-choice tree. [pcc:unverified]

Why the pruning rules fit the target

The initial E>R guard admits only remaining<=R if the exponent sum fits. Each committed push explicitly requires remaining<=R-row-1; a pop grows the row capacity by one and restores at most one consumed exponent. These give the proposed remaining<=R-row induction, independently of assuming the Nov5 driver-level E<=R argument. [pcc:unverified]

At a non-leaf let m be remaining before trying a candidate and k=R-row>=1. The capacity invariant gives m<=k. A mode choice has m>=1 and changes it to m-1<=k-1, so it cannot fail the degree test. A constant leaves m unchanged and fails exactly when m=k. Thus the source's conditional mode restoration inside the degree-rejection block is defensive for this admitted-state model. This is a written reachability argument, not an executed branch test or a checked theorem; no branch or guard is removed. [pcc:unverified]

After a tentative choice at row r, every remaining affine row contributes degree0 or one incoming variable. If total residual degree exceeds R-r-1, no suffix can supply that exponent vector. Rejecting a mode with left_j=0 likewise discards only over-target exponents. Both arguments require nonnegative formal exponents and affine row factors, not nonzero coefficients. [pcc:unverified]

Zero testing applies to the current factor. The implementation does not prune merely because prefix[row] is zero. Nonzero factors may multiply to zero in the creation ring; such branches can still be traversed and add zero. A cost argument must retain their work. Zero-factor pruning and the final zero-result shortcuts use annihilation, without dividing or cancelling any coefficient. [pcc:unverified]

At a leaf, nonnegative left_j values with remaining=0 mean the target beta was used exactly. Ordered row choices are accumulated individually: matching coefficient values do not justify deduplication or a factorial adjustment. The finite row-choice tree and monotonically advanced row cursors supply the proposed termination argument for successful local execution. This does not establish the general entry-work or recursion bound. [pcc:unverified]

Hand-derived trace and edge controls

The following is a symbolic source walkthrough, not an executed test. With two rows, one incoming label and beta=1:

Prefix / candidate	Effect
Row0 chooses constant; row1 chooses incoming	First eligible leaf uses the incoming label at row1
Return from that leaf	Pop row1's committed choice and restore its one exponent
Exhaust row1 under constant prefix	Pop row0's constant; no exponent restoration
Row0 chooses incoming; row1 tries constant	Second eligible leaf uses the incoming label at row0
Row1 next tries incoming	left=0, so the extra use is skipped
Return to root exhaustion	Both ordered leaves are closed; no extra leaf is introduced

For a degree-rejection walkthrough use beta=2 instead: at each of two rows the constant choice leaves too much degree for the suffix and is rejected; the mode choice consumes one unit and fits. No speculative mode restoration is reached under the capacity invariant. Both parent-pop sites keep their separate role. Equal row coefficients still leave indexed leaf positions. [pcc:unverified]

At R=0 and E=0, the leaf case reads prefix[0]=1 and breaks without indexing empty choices/chosen arrays, then applies outgoing factors. At R=0,E>0 the degree guard returns zero. With q=0, only constant choices remain. Records with zero target exponent are skipped by the left=0 rule; positivity of incoming exponents is needed for the earlier q<=E bound, not for this semantic model. Keep all existing guards. [pcc:unverified]

Errors, integer state and immutable precision

The relation above concerns successful transitions and Ok results. A failing ring multiplication can occur after a tentative decrement but before prefix and chosen are committed. Result::Err exits the function; the local DFS arrays are discarded, with no suspended state or partial coefficient returned. Meter charges persist as diagnostics. Do not claim every error-state array satisfies the active-prefix predicate or that an error is a complete count. [pcc:unverified]

The shape/width contract also matters. choices[row]+1 needs q+1 to fit; row+1 and prefix allocation need R+1. Exponent subtraction occurs only after left_j>0. A parent pop is guarded by row>0, while R-row-1 is evaluated only inside row<R. Restoring an actually consumed unit keeps counters at most their initial beta/E. The unverified Nov5 E<=R caller contract can discharge the incoming-sum width condition if refined; it is not silently assumed here. [pcc:unverified]

Outgoing sums are accumulated modulo the stored creation modulus, embedded as ring integers and exponentiated with the source's repeated-squaring loop. Exponent0 returns one, including a zero base; actual child construction omits zero-exponent records. Only after the complete packet entry does Tensor::entry lift coordinates to the requested precision. The source rejects decreased precision or a packet/creation mismatch. A canonical lift is not assumed to commute with the earlier arithmetic. [pcc:unverified]

Review status and next discriminating step

This packet supplies the signed semantic target, active-prefix/frontier predicates and a source argument for each trial, prune, commit and restoration. The missing checked bridge is a faithful coefficient/choice-tree model plus an imperative-state refinement covering these exact code sites. Lean/lake are unavailable; no theorem module, transitive axiom audit or polynomial identity certificate is presented. Standard D.4.6 semantics are being audited, not promoted as a newly certified algebraic identity. [pcc:unverified]

Next, check that refinement with the stated commutative-ring and machine-state preconditions. Then combine entry preparation, all attempted DFS choices, outgoing powers and operand widths with the ancestor-retention ledger. Kernel coefficients, packing/tuple correspondence and the full general audit remain explicit dependencies.

The upstream bounded controls (../prime_ic_elimination_20261101/RANK1_CODE_CONTROLS.md) and new mutation tests were read as historical/source evidence, not rerun or substituted for this refinement. No scientific experiment, benchmark, challenge solve or cloud resource ran. Earlier reports, tests, gates and dependencies remain unchanged. Canonical leaderboard, progress, curve and performance graphs receive no new measurement or transfer edge.

The PDF (report.pdf), editable diagram (dependency.dot), renderer (render_report.py) and validation note (VALIDATION.md) accompany the packet. The general-audit problem remains open.